DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

Neural Network Lab - Assessment Report

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Student Information

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Overall Result

26/28 (92.9%) Grade: A

Score by Section

Section	Score	Pct
Activation Functions	4/5	80.0%
Weight Initialization	3/3	100.0%
Loss Functions	3/3	100.0%
Optimizers	3/4	75.0%
Backpropagation	3/3	100.0%
Regularization	4/4	100.0%
Batch Normalization	2/2	100.0%
Architecture Design	4/4	100.0%

Questions to Review

Q03. [Activation Functions]

Why is Tanh preferred over Sigmoid for hidden layers?

Tanh outputs range (-1, 1) centered around 0. Sigmoid outputs (0, 1) are always positive, causing zig-zag gradient updates. Zero-centering avoids this inefficiency.

Q14. [Optimizers]

Which optimizer typically converges fastest on small datasets?

Adam adapts the learning rate per-parameter using gradient history, making it robust to different gradient scales.